Developer Collaboration with Git and GitHub

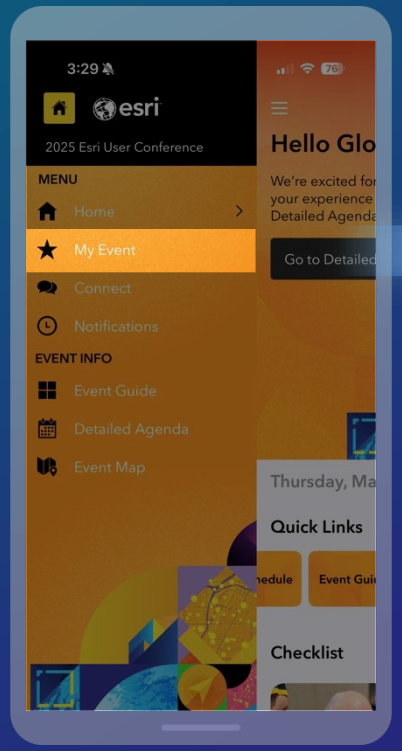
Nate Bedortha and Courtney Yatteau

ESRI DEVELOPER & TECHNOLOGY SUMMIT

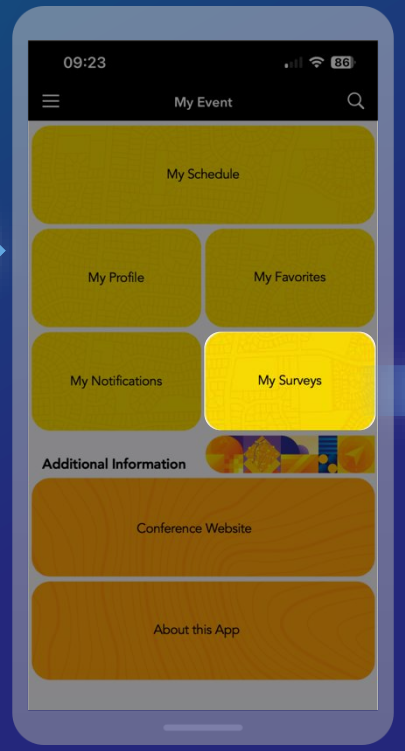
Copyright © 2026 Esri.
All rights reserved.

Please share your feedback in the app

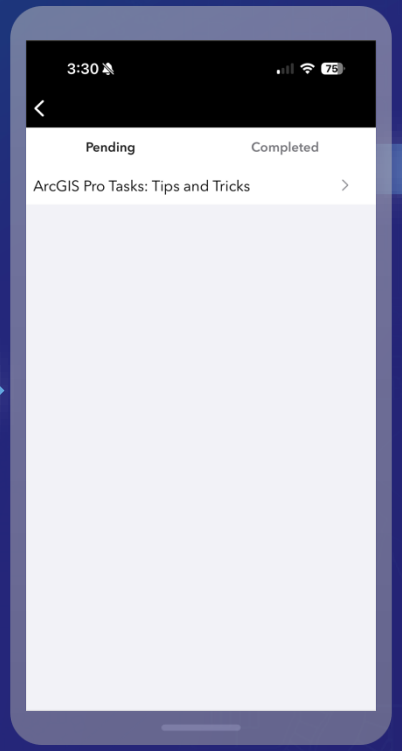
Select “My Event” in the event app main menu



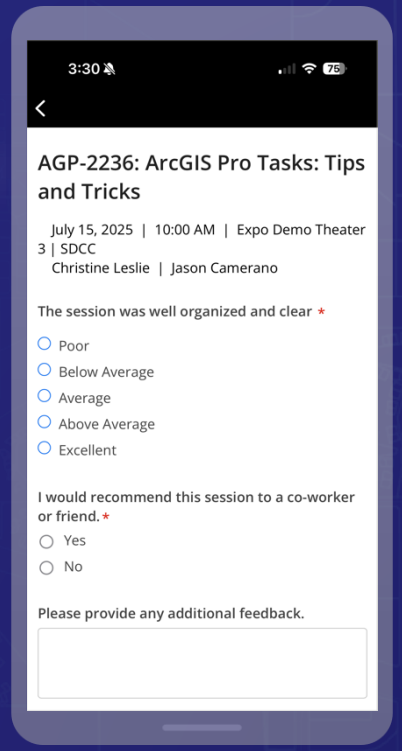
Select “My Surveys”



Select the desired survey



Complete the survey



Courtney Yatteau

Developer Advocate, Esri

c_yatteau

c_yatteau

cyatteau

courtneyyatteau

cyatteau.bsky.social

Nate Bedortha

Software Development Engineer, Esri

oknoway

Agenda

What we'll cover

🧠 Mental model

- draft ➔ review ➔ publish

👩🏻‍💻 Live demo

- branch ➔ commit ➔ pull request ➔ review ➔ merge

📦 Real Esri repos

- Issues, pull requests, contributing

❓ Questions

Draft first. Publish second.

A simple safety pattern for teamwork

Why teams use version control

So collaboration feels safe.

Shared work is easier when changes are visible, reversible, and reviewable.

What version control gives teams

Three practical benefits

Undo mistakes

Track what changed

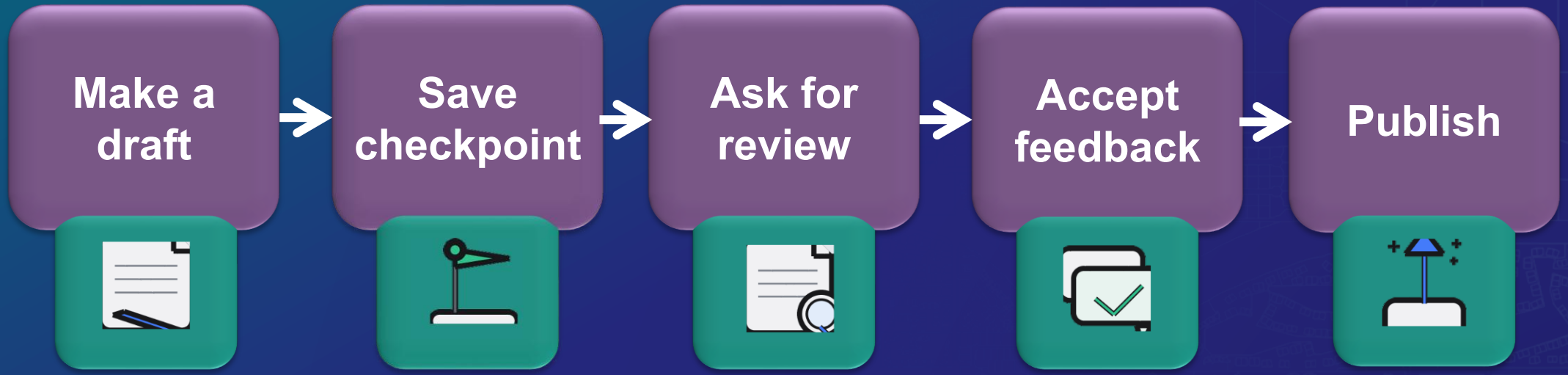
Work in parallel

Main should be boring.

Changes go through pull requests

Five steps you'll see today

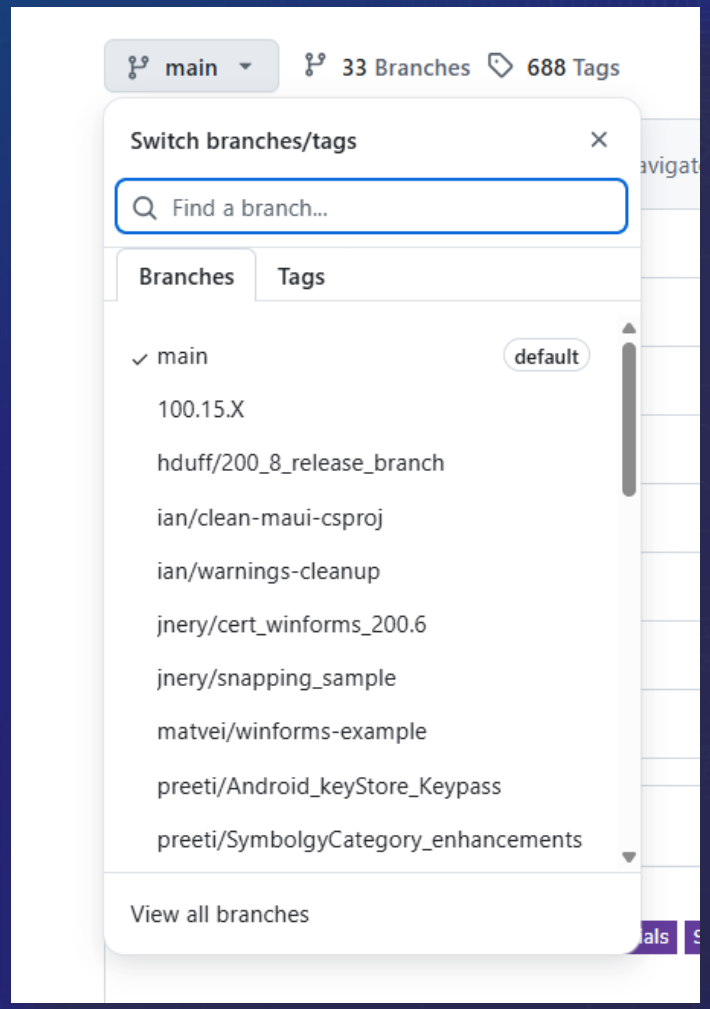
Same pattern for code, docs, etc.



Step 1: Branch

Create a safe draft space

- Keeps main stable
- Lets you work without affecting others
- Easy to throw away if needed

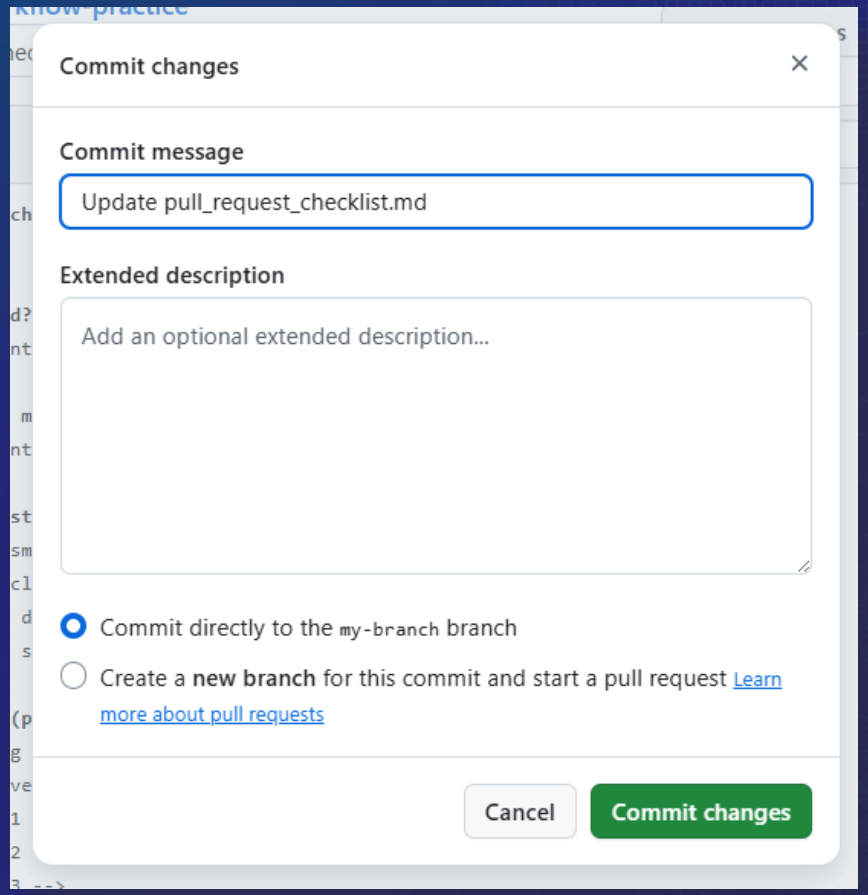


Draft **Branch**	Checkpoint	Ask for review	Feedback	Publish

Step 2: Commit

Save a checkpoint

- Records changes to files
- Has a unique ID
- Includes a short message

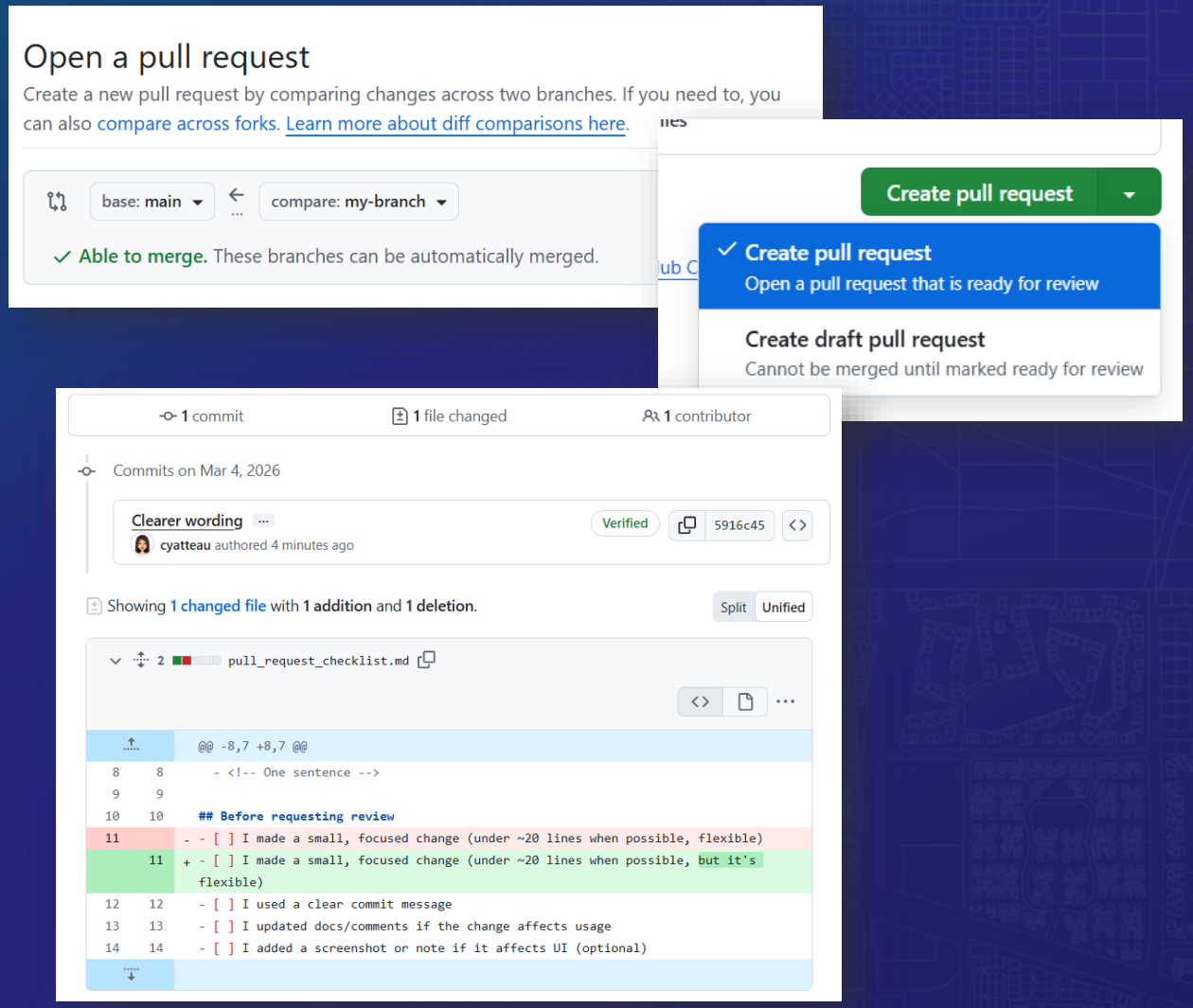


Draft **Branch** → Checkpoint **Commit** | Ask for review | Feedback | Publish

Step 3: Pull request

Ask for review before publishing

- Propose a merge into main
- Holds context, discussion, and review
- Shows the diff in ‘Files changed’

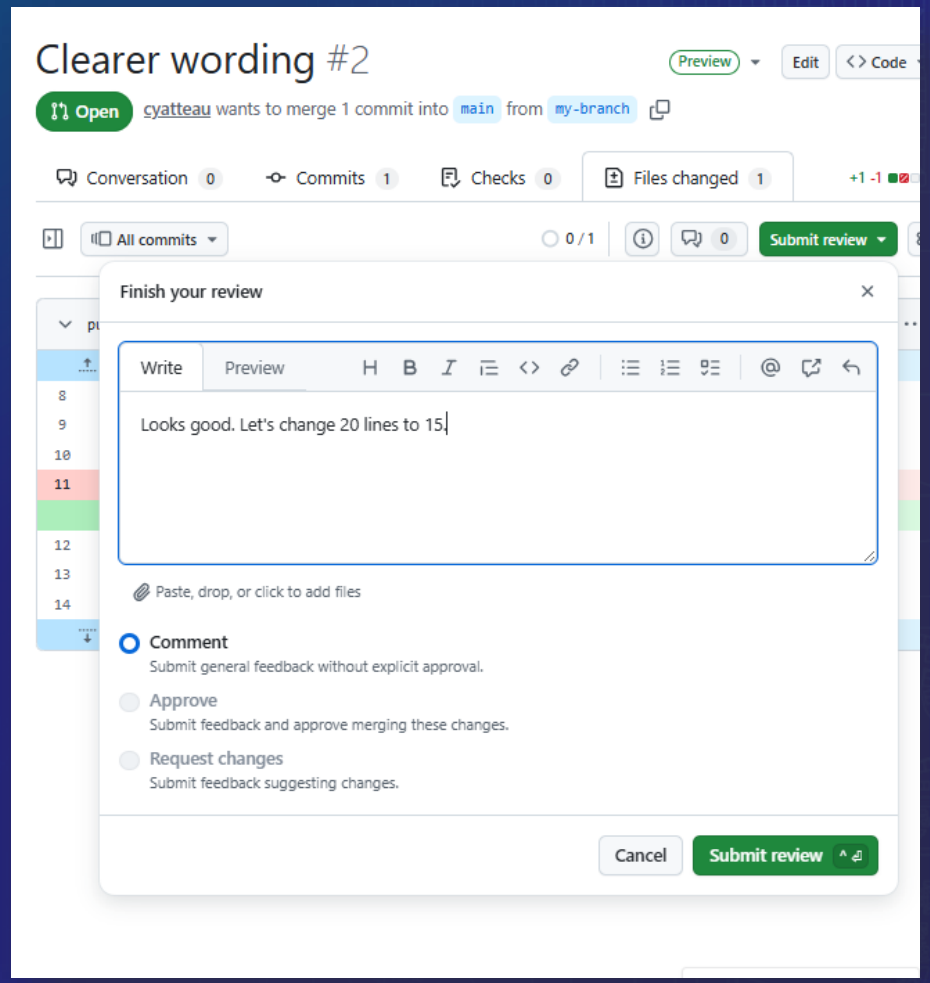


Draft **Branch** → Checkpoint **Commit** → Ask for review **Pull request** | Feedback | Publish

Step 4: Review

Feedback before it hits main

- Comment on specific lines
- Approve or request changes
- One file at a time is the easiest way

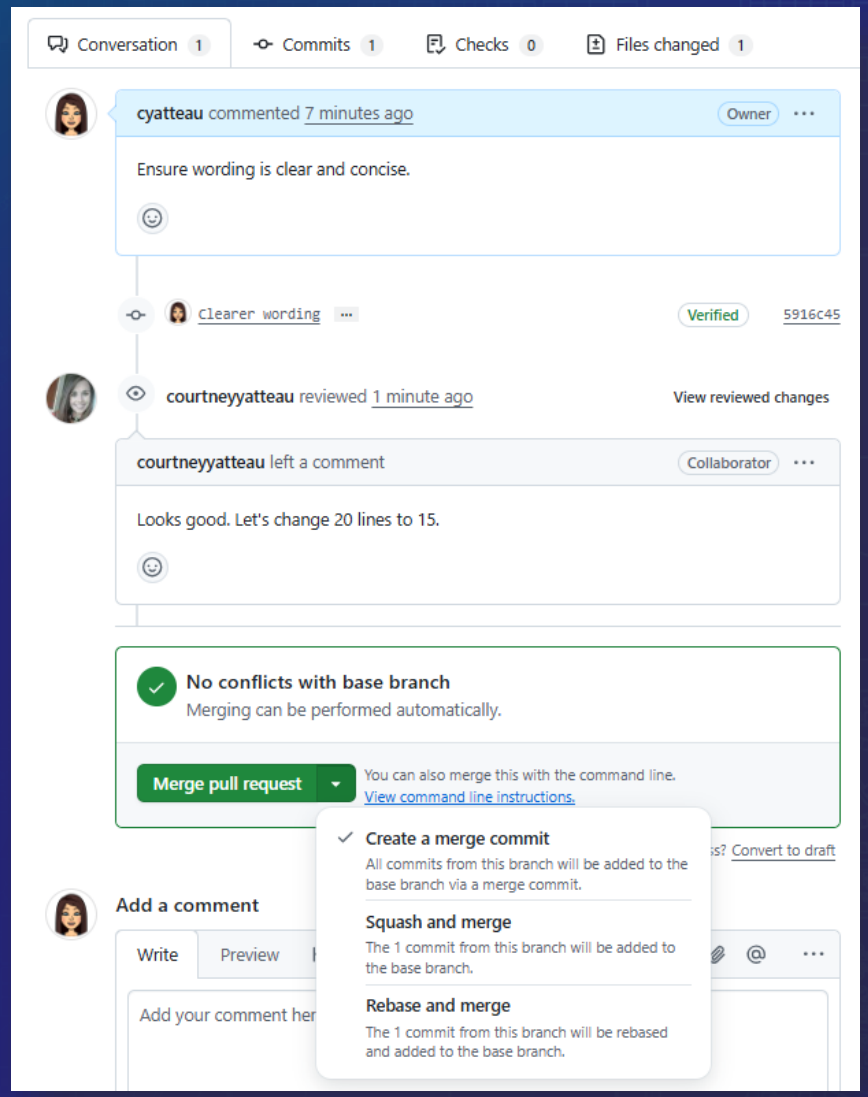


Draft **Branch** → Checkpoint **Commit** → Ask for review **Pull request** → Feedback **Review** | Publish

Step 5: Merge

Publish to the default branch

- Moves changes into main
- PR keeps comment + commit history
- This is the publish moment



Draft **Branch** → Checkpoint **Commit** → Ask for review **Pull request** → Feedback **Review** → Publish **Merge**

Three merge options on GitHub

Same result, different history

Merge commit

- Keeps all branch commits
- Adds a merge commit

Squash and merge

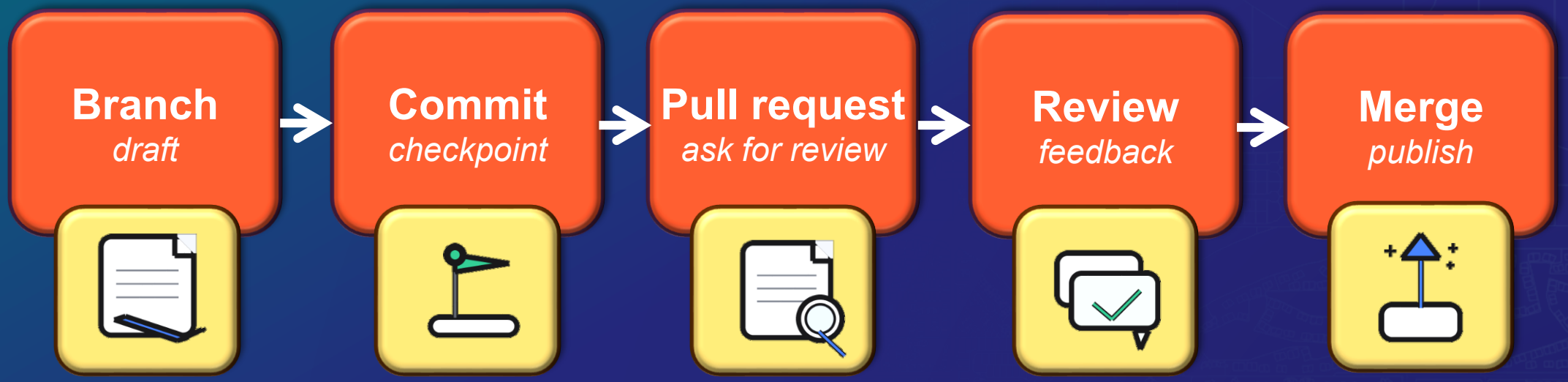
- Combines the PR into one commit
- Keeps main cleaner

Rebase and merge

- Re-applies commits onto the end of main
- Keeps history in one straight line

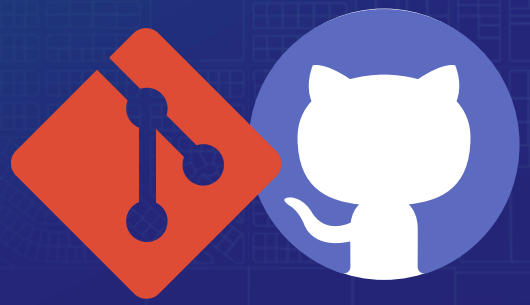
GitHub labels for those steps

Same story, official names in the UI



What happens ‘under the hood’

Git records history, GitHub hosts and coordinates

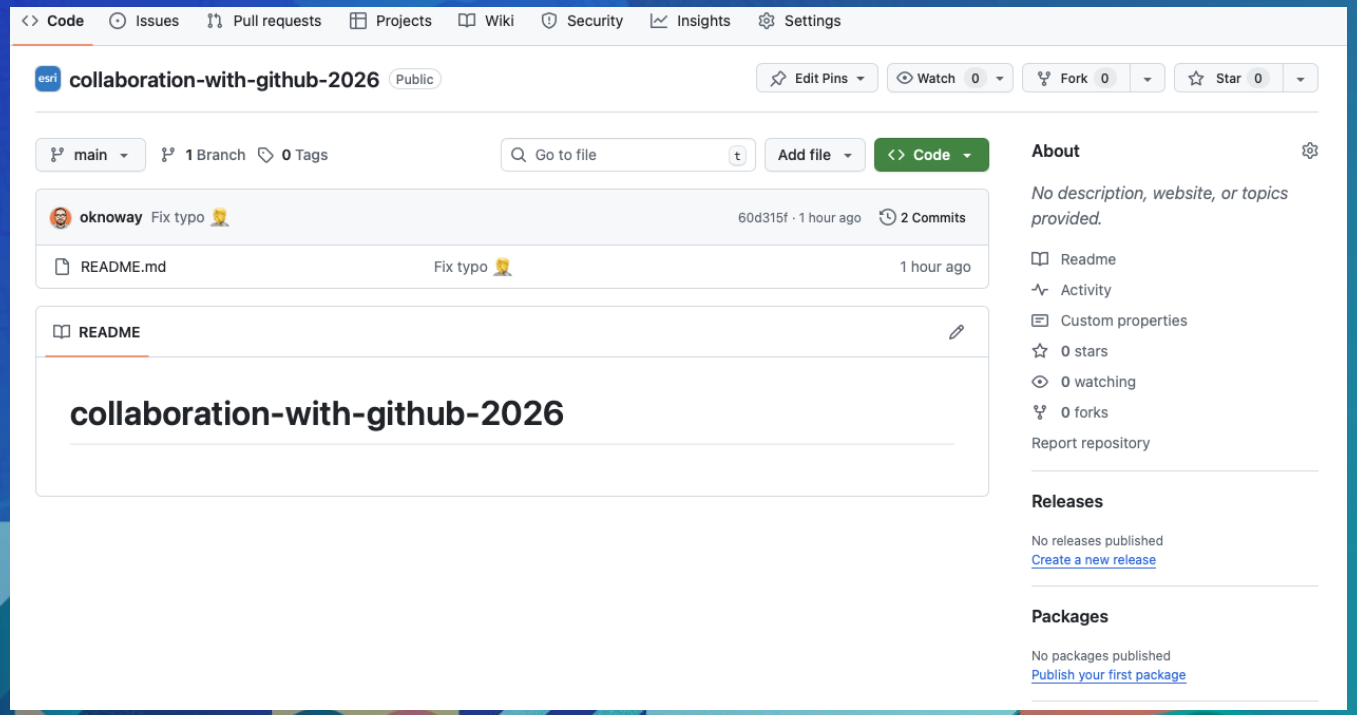


Git

- Records change history
- Stores checkpoints (commits)
- Supports draft copies (branches)

GitHub

- Hosts repo online
- Adds collaboration (PRs, review, issues)



Live Demo

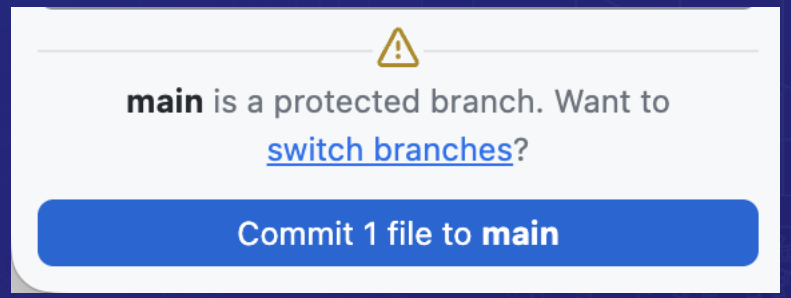
Let’s run the loop in the browser

When merge is blocked

GitHub is protecting your default branch

Scenario 1, Protected branch

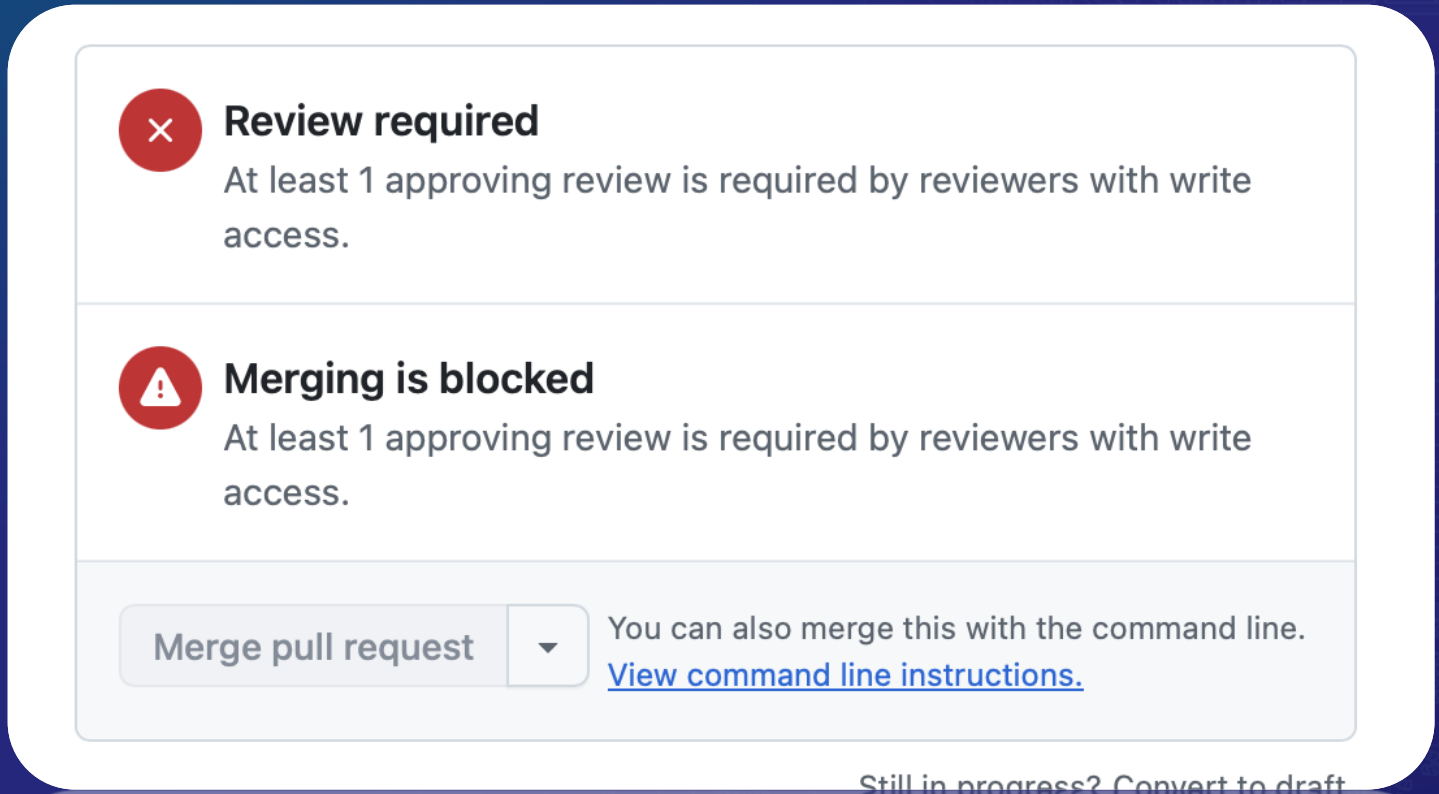
- You can’t push to main directly
- Changes must come through a pull request
- Rules can require reviews



Scenario 2, Reviews required

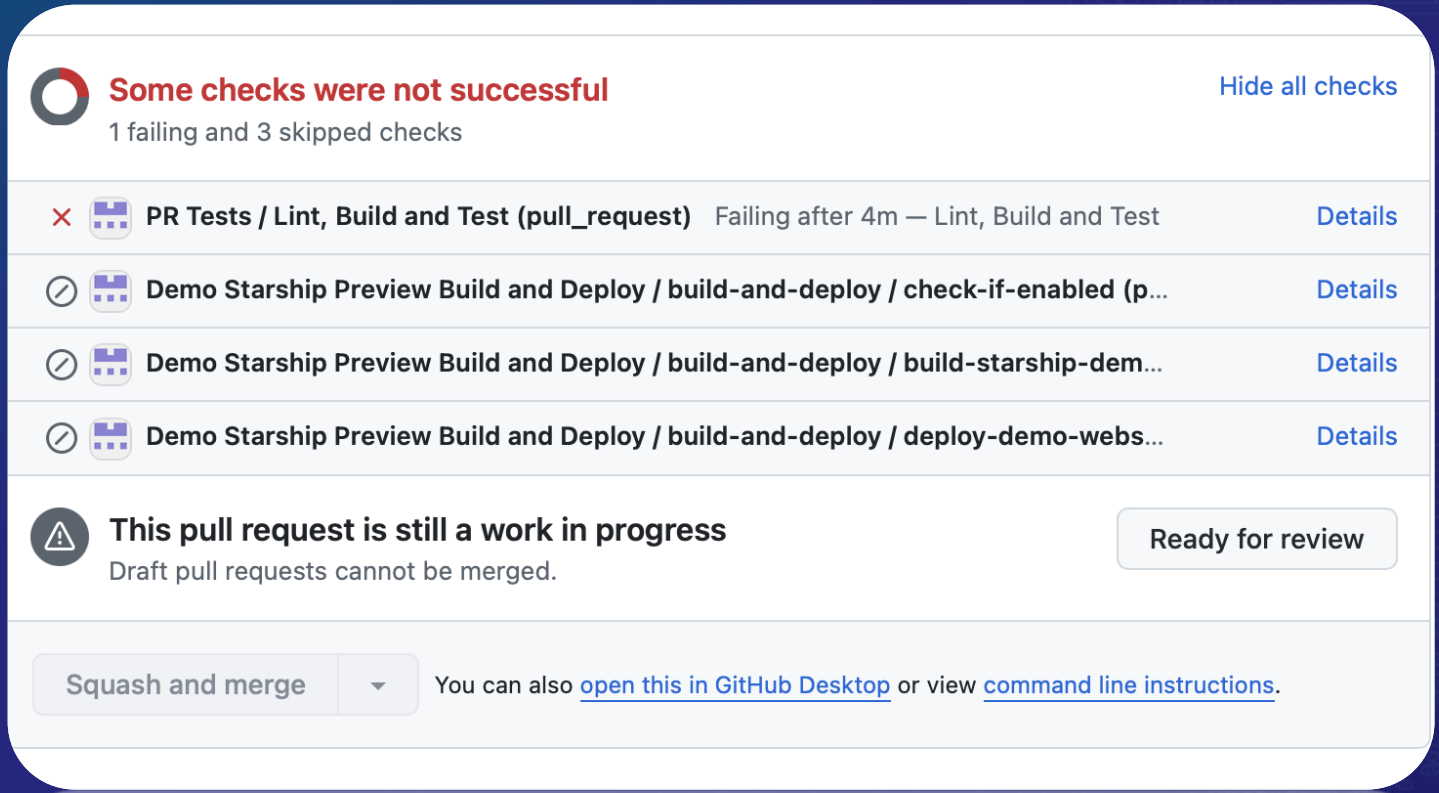
Make it easy to reproduce and respond

- Needs approval before merge
- ‘Request changes’ blocks merge until addressed
- New commits may dismiss prior approvals



Scenario 3, Checks required

- Status checks must pass before merge
- Red X means something failed
- Look in the PR ‘Checks’ section for details



You can contribute

Start small, still valuable

📝 Improve docs

🐞 File clear issues

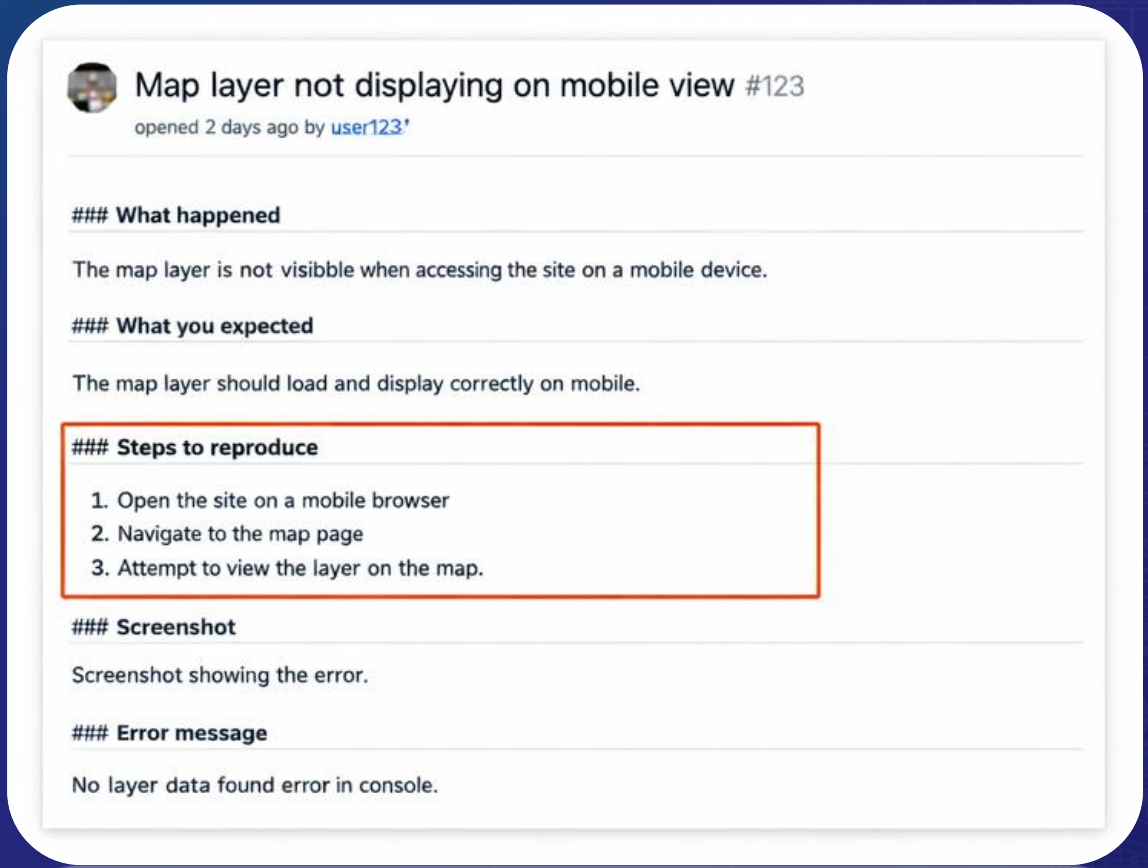
👀 Review pull requests

🔧 Submit a small PR

What makes a great Issue

Make it easy to reproduce and respond

- ✅ What happened
- 🎯 What you expected
- 🧭 Steps to reproduce
- 📎 Screenshot or error text



Map layer not displaying on mobile view #123

opened 2 days ago by user123'

What happened

The map layer is not visibble when accessing the site on a mobile device.

What you expected

The map layer should load and display correctly on mobile.

Steps to reproduce

1. Open the site on a mobile browser
2. Navigate to the map page
3. Attempt to view the layer on the map.

Screenshot

Screenshot showing the error.

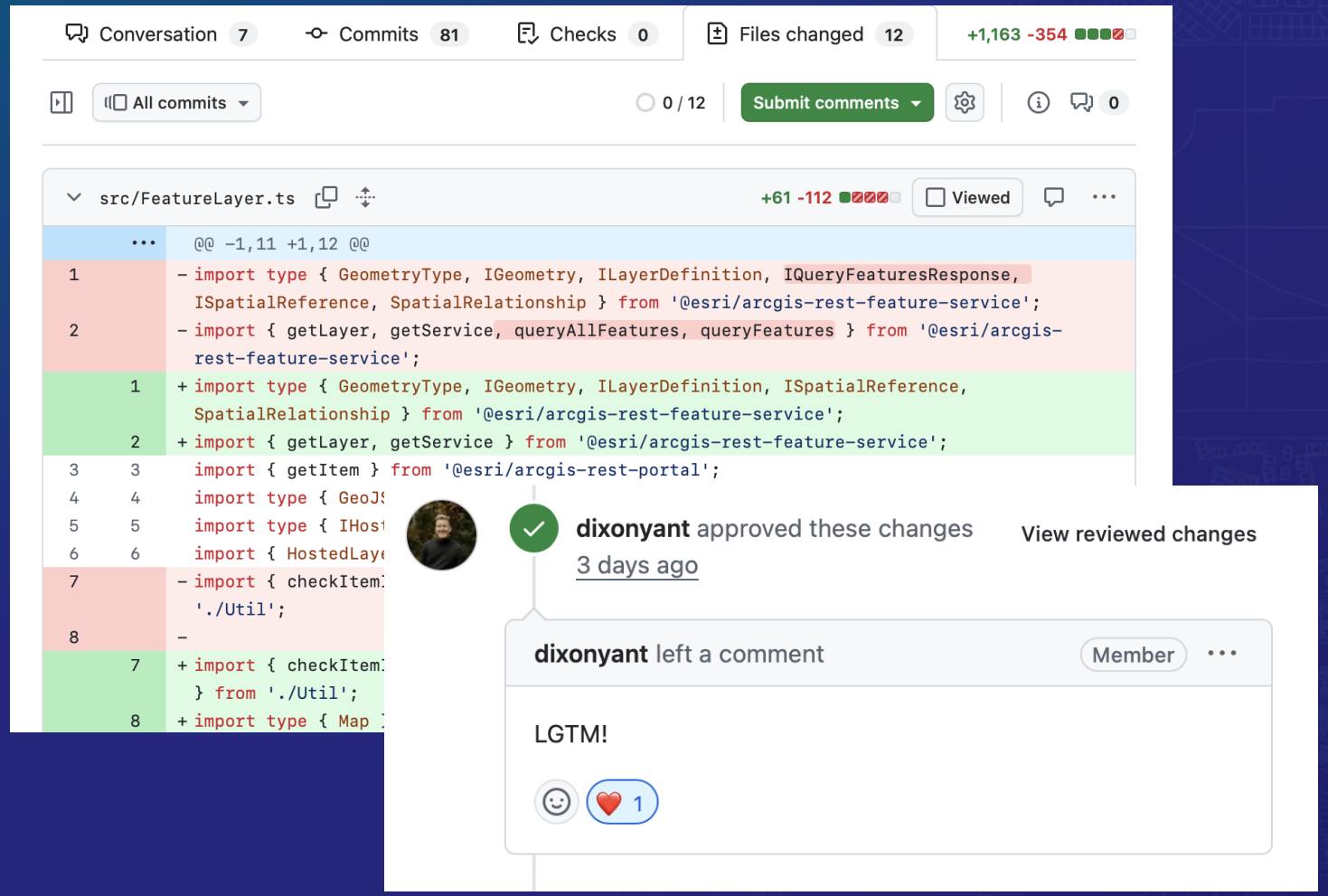
Error message

No layer data found error in console.

Pull requests are changes in motion

Context, diff, and decision

- 💬 Conversation = why
- 📄 Files changed = what
- ✅ Merge = publish



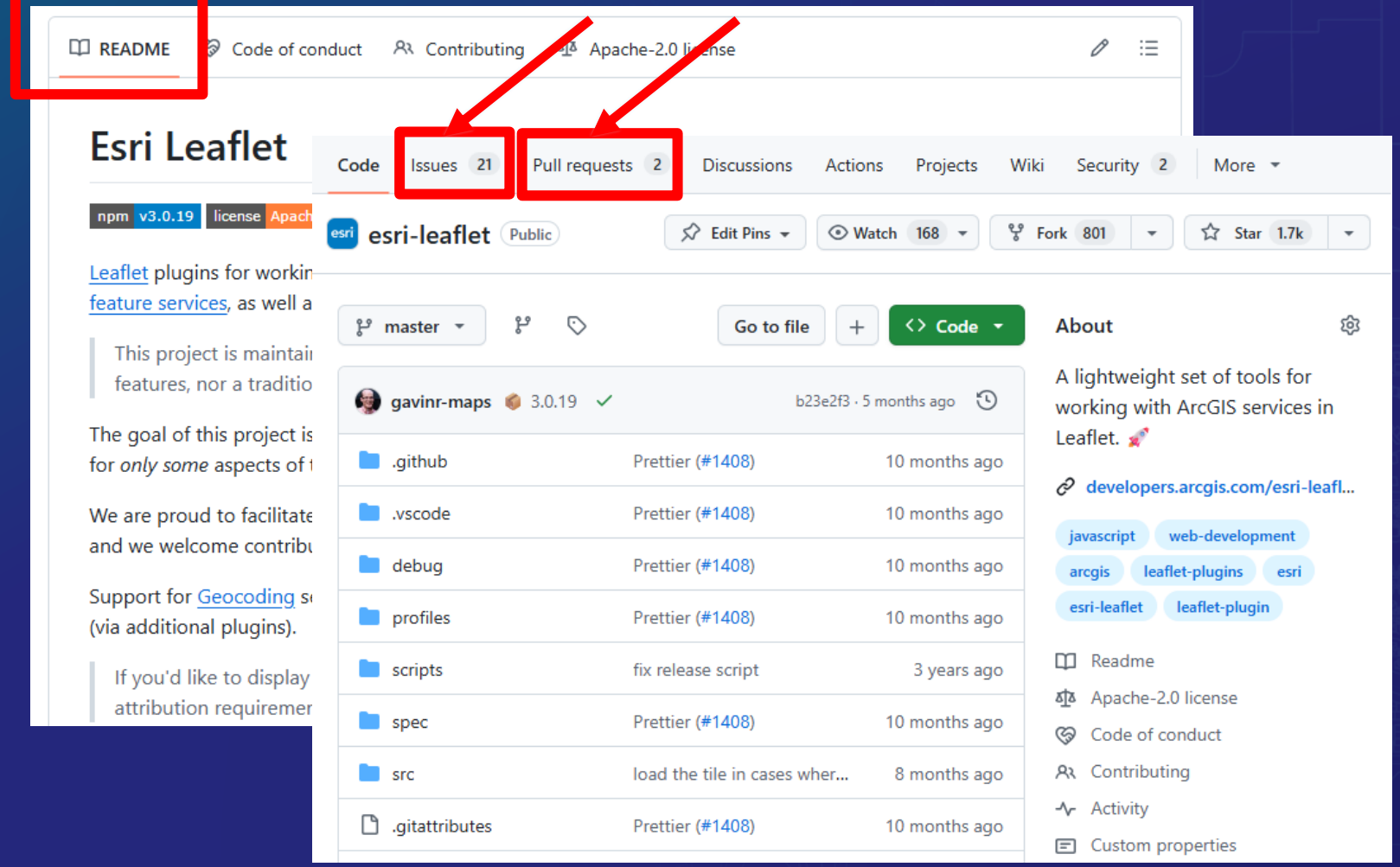
Real Esri repos

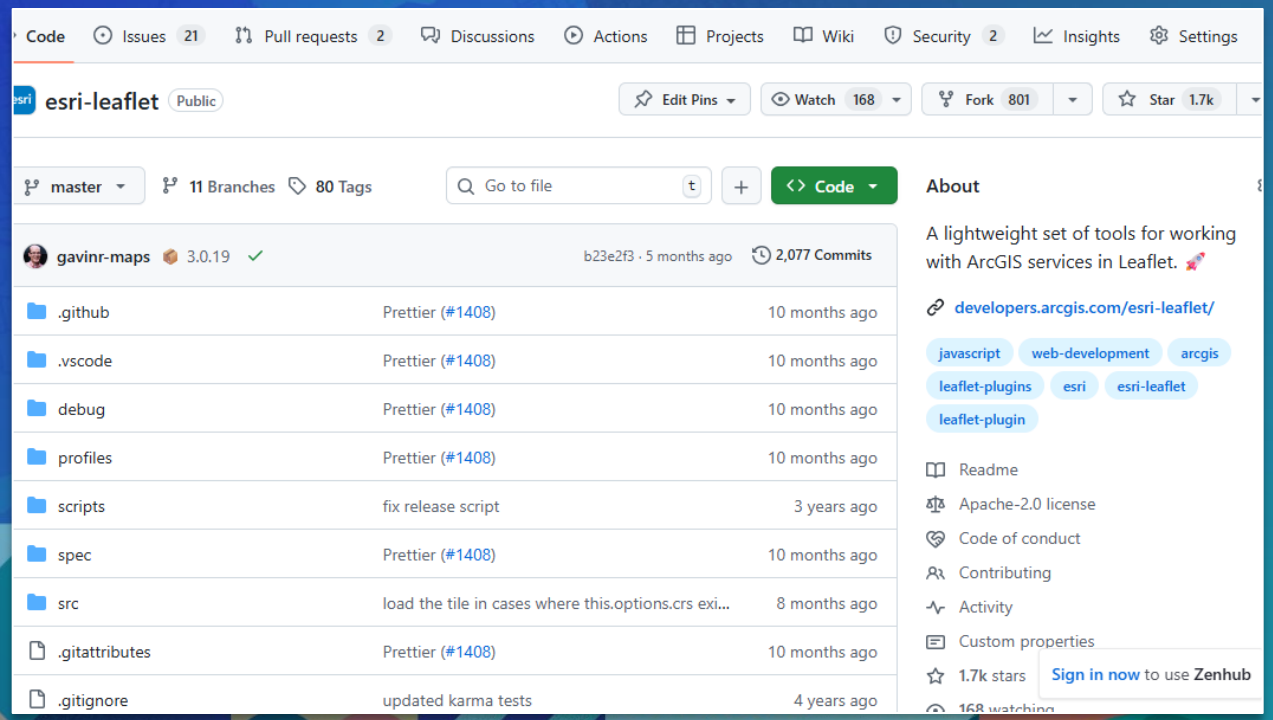
Same workflow, real projects

How to read a repo fast

Start with intent, then activity

- README
- Issues
- Pull requests



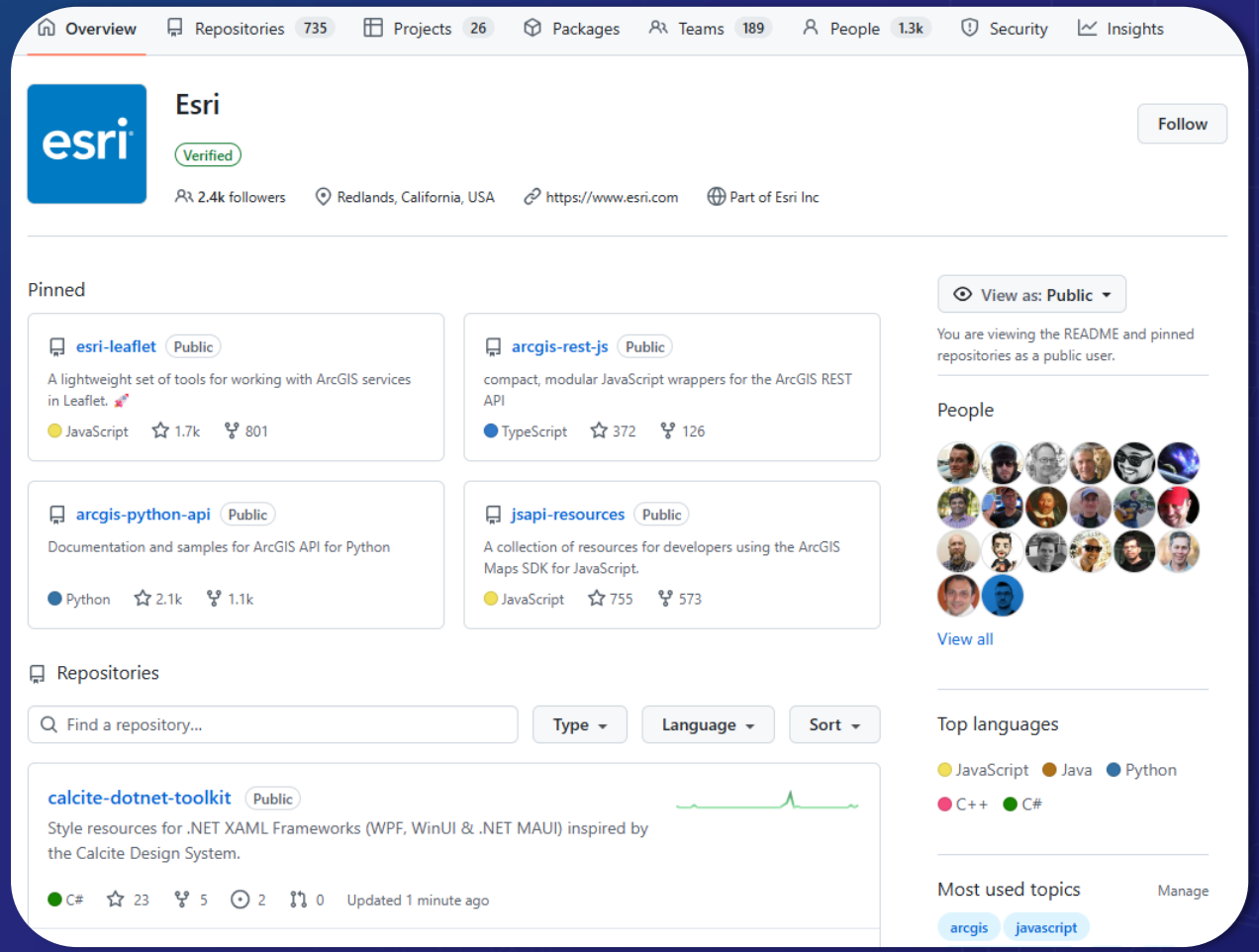


Live repo walkthrough

Esri Open Source on GitHub

Real projects, real collaboration

- 700+ public repositories
- Used by developers worldwide
- Community contributions welcome

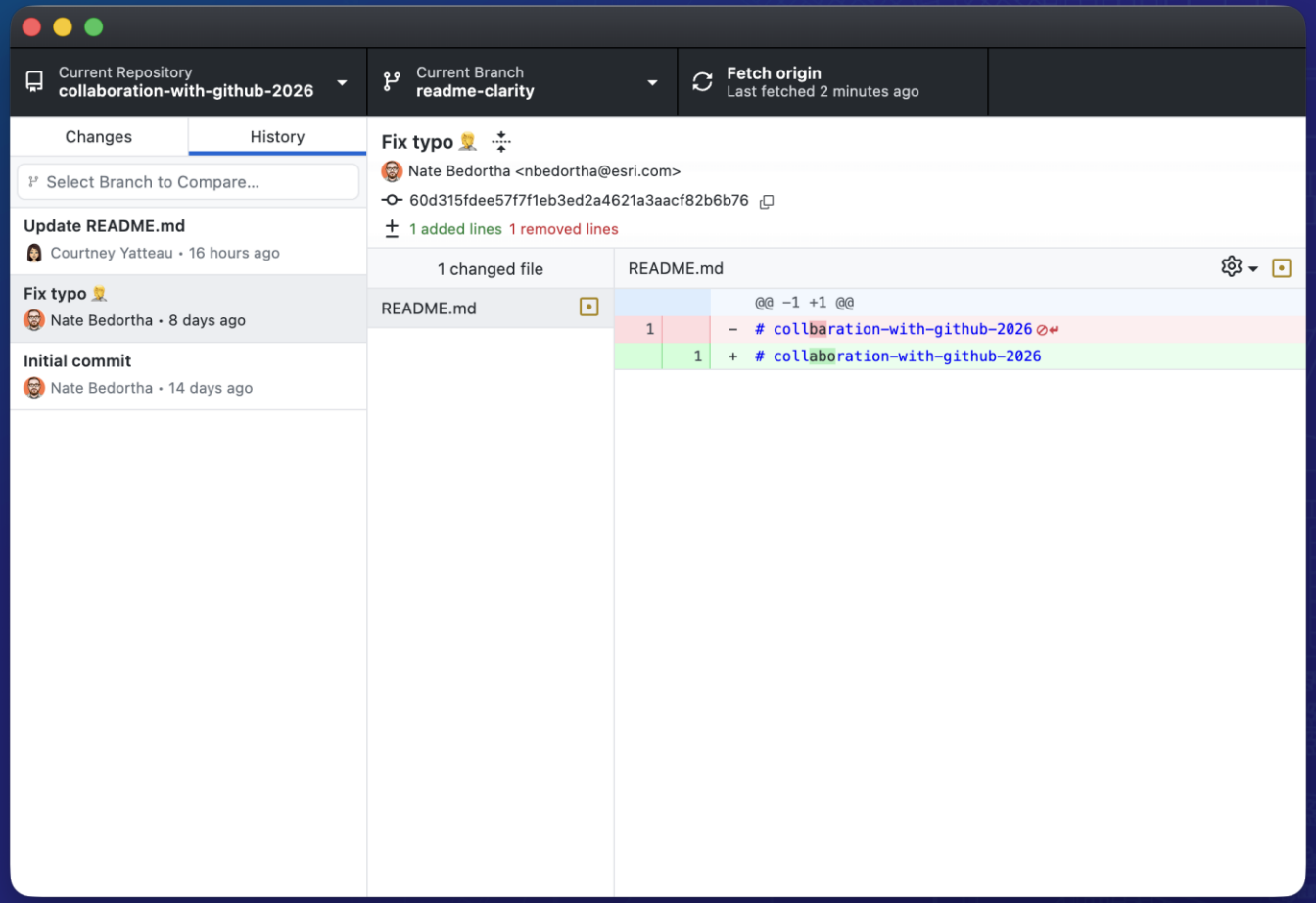


GitHub Desktop

"Experience Git without the struggle"

- GUI for branching, committing, opening pull requests, and merging.
- Easily compare diffs and code changes before you commit.
- Browse commit history, append or revert old commits, stash changes for later.

desktop.github.com



Hacktoberfest & Community Contributions

Open source participation at Esri

- Participate annually in Hacktoberfest
- Label beginner-friendly issues
- Encourage first-time contributors

Resources

Esri's open-source GitHub repos: esri.github.io

GitHub Skills: Intro to Git
github.com/skills/introduction-to-github

GitHub Docs: Hello World
docs.github.com/en/get-started/start-your-journey/hello-world

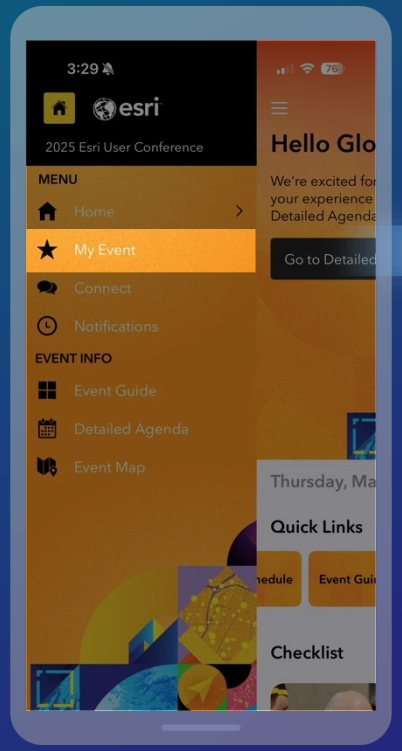
Git SCM
git-scm.com

Wizard Zines: How Git Works
wizardzines.com/zines/git/

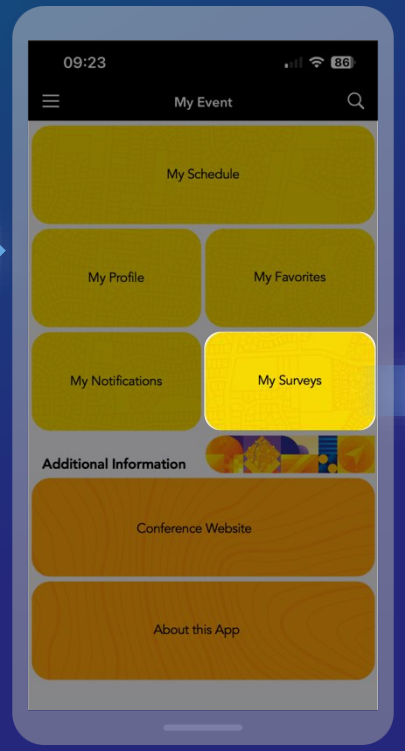
Haunted Places Repo
github.com/cyatteau/haunted-places-map-assistant

Please share your feedback in the app

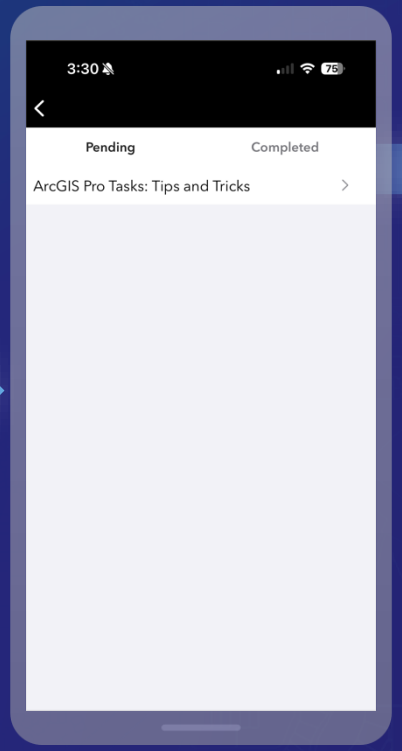
Select “My Event” in the event app main menu



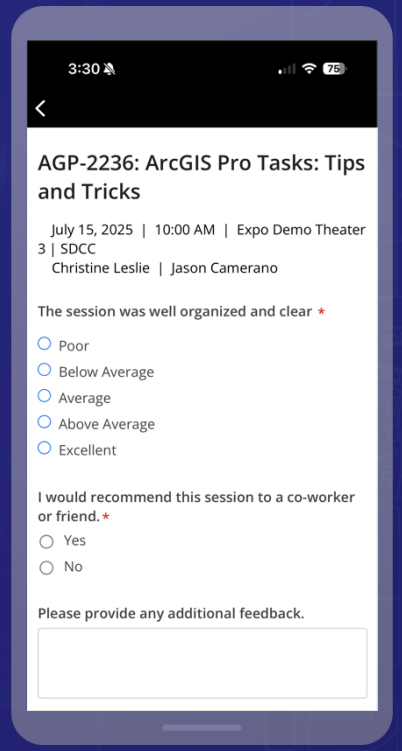
Select “My Surveys”



Select the desired survey



Complete the survey



Connect with us on social

Join the conversation using #EsriDevTech2026

- x.com/EsriDevs
- x.com/EsriDevEvents
- youtube.com/@EsriDevs
- links.esri.com/DevVideos
- github.com/Esri
- github.com/EsriDevEvents
- links.esri.com/EsriDevCommunity
- linkedin.com/showcase/esri-developers

Questions

esri®

THE SCIENCE OF WHERE®

Copyright © 2026 Esri. All rights reserved.